

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

Iteration 1

Dataset:

Loves Popcorn	Favorite Color	Age	Loves Trail
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step1: Calculate (Initial) Prediction

Loves Popcorn	Favorite Color	Age	Loves Trot2
1.6	Blue	12	Yes 
1.6	Green	87	Yes 
1.5	Blue	44	No 
1.8	Red	19	No 
1.5	Green	32	Yes 
1.4	Blue	14	Yes 

Step2: Calculate Residual

Lowes Popcorn	Favorite Color	Age	Low Tim
1.6	Blue	12	Ye
1.6	Green	87	Ye
1.5	Blue	44	Ni
1.8	Red	19	Ni
1.5	Green	32	Ye
1.4	Blue	14	Ye

(Initial) Prediction	Residual
0.7	0.3
0.7	0.3
0.7	-0.7
0.7	-0.7
0.7	0.3

Step3 Build a tree to predict the residuals:

Loves Popcorn	Favorite Color	Age	Loves Trot2	Res after init guess
1.6	Blue	12	Yes	0.3
1.6	Green	87	Yes	0.3
1.5	Blue	44	No	-0.7
1.8	Red	19	No	-0.7
1.5	Green	32	Yes	0.3
1.4	Blue	14	Yes	0.3

Color = Best

```

graph TD
    Root[Gender = Male] -- Yes --> Leaf1((4))
    Leaf1 --- Val1[-3.3]
    Root -- No --> Node1[Age > 37]
    Node1 -- Yes --> Leaf2((23))
    Node1 -- No --> Leaf3((616))
  
```

$$= \frac{\text{Sum(Residuals)}}{\text{Sum (Previous probability) \times (1 - Previous probability)}}$$

Iteration 2

Tree after 1st iteration

```

graph TD
    Root[Color = Red] -- Yes --> L[-0.7]
    Root -- No --> R[Age > 37]
    R -- Yes --> RL[0.3]
    R -- No --> RR[0.3, 0.3, 0.3]
  
```

Step1: Calculate (updated) Prediction

Loves Popcorn	Favorite Color	Age	Loves Trotz
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Calculate Residual

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step3: Build a tree to predict the residuals

Loves Popcorn	Favorite Color	Age	Loves Trot2	residuals
1.6	Blue	12	Yes	0.1
1.6	Green	87	Yes	0.1
1.5	Blue	44	No	-0.5
1.8	Red	19	No	-0.3
1.5	Green	32	Yes	0.1
1.4	Blue	14	Yes	-0.1

```

graph TD
    A["Age <= 56"] -- No --> B((2))
    A -- Yes --> C["Age > 37"]
    C -- Yes --> D((3))
    C -- No --> E((4))
  
```

$$= \frac{\text{Sum(Residuals)}}{\text{sum}[(\text{Previous probability}) \cdot (1 - \text{Previous probability})]}$$
 $109(1-0.9)$

Iteration 3

Tree after 2nd Iteration

A decision tree diagram for predicting 'Age'. The root node is 'Age < 66'. If 'No', it leads to a leaf node with value '0.5', which has a handwritten green circle around it and a green checkmark below it. If 'Yes', it leads to another node 'Age > 37'. From 'Age > 37', if 'Yes', it leads to a leaf node with value '-0.5', which has a handwritten green circle around it and a green checkmark below it. If 'No', it leads to a leaf node with value '0.1,-0.1,0.1,0.1', which has a green checkmark below it.

```

graph TD
    Root[Age < 66] -- No --> L1[0.5]
    Root -- Yes --> L2[Age > 37]
    L2 -- Yes --> L3[-0.5]
    L2 -- No --> L4[0.1,-0.1,0.1,0.1]
  
```

Step1: Calculate (updated) Prediction

Lowes Popcorn	Favorite Color	Age	Lowes Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Calculate Residual

Loves Popcorn	Favorite Color	Age	Loves Trail2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

Step2: Build a tree to predict the residuals

Loves Popcorn	Favorite Color	Age	Loves Troll2
1.6	Blue	12	Yes
1.6	Green	87	Yes
1.5	Blue	44	No
1.8	Red	19	No
1.5	Green	32	Yes
1.4	Blue	14	Yes

log of odds = $\beta_0 + \beta_1 x$ Prediction of Residual from tree-1 $\rightarrow$ $\beta_0 + \beta_1 x$ Prediction of Residual from tree-2
 $\rightarrow \mathcal{L}(\log \text{ of odds})$

Prediction from Trained model

```

graph TD
    Root[Color = Red] -- Yes --> L1[-0.7  
-3.3]
    Root -- No --> Node1[Age > 37]
    Node1 -- Yes --> L2[0.3, -0.7  
-1]
    Node1 -- No --> L3[0.3, 0.3, 0.3  
1.4]
  
```

Tree after 2nd Iteration

```

graph TD
    A[Age <= 66] -- No --> B[0.5  
2]
    A -- Yes --> C[Age > 37]
    C -- No --> D["0.1, 0.1, 0.1, 0.1  
0.6"]
    C -- Yes --> E["-0.5  
-2"]
  
```

New data:

Loves Popcorn	Favorite Color	Age	Loves Tru2
1.7	Blue	44	? $\rightarrow 1.9$ (Y)

 $\Delta \rightarrow N\bar{D}$

Log of odds =

$1 + e^{-2} \rightarrow 0.87 > 0.5$
 $0.35 < 0.5$

X